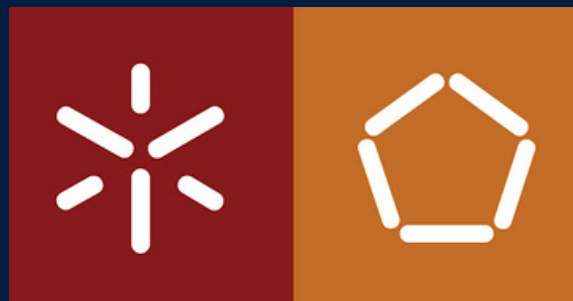


LAW TALK

Data Mining



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24/05/2024

INDEX

01

Introduction

02

Architecture

03

Data Sources

04

Data Extraction
and Database

05

Graphical Interface

06

Result Analysis

INTRODUCTION

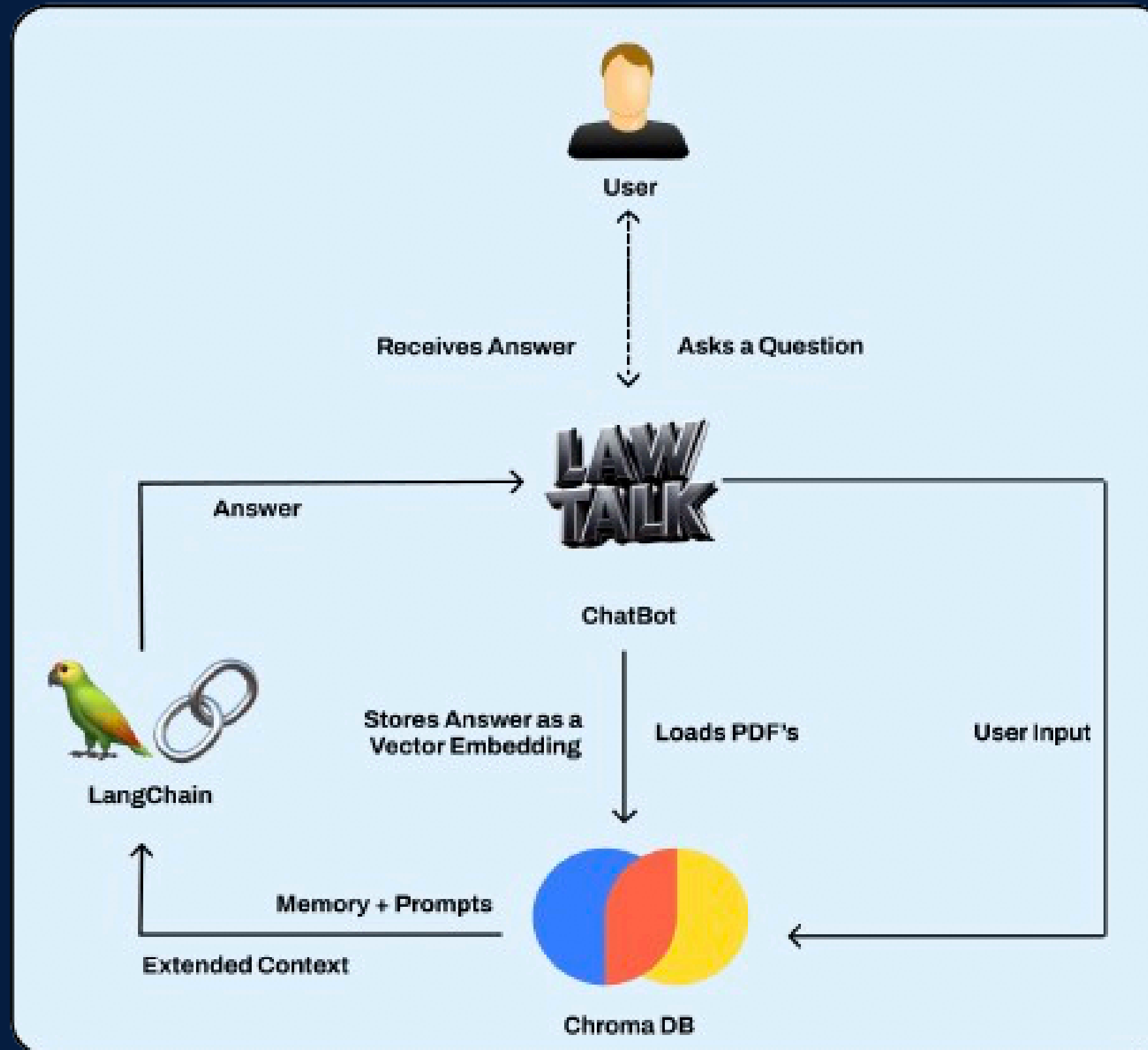
CONTEXTUALIZATION

- 01 Chatbot capable of solving practical cases corresponding to crimes, providing information about the legal consequences to be applied to the offender.
- 02 Chatbot capable of citing the articles on which it relied to solve the practical cases.

OBJECTIVES

- 01 Practical ChatBot that answers in Natural Language
- 02 Customization in terms of LLM model options and embeddings.

ARCHITECTURE



DEVELOPMENT PLAN

RESOURCES

01 Language – Python

02 Framework – Langchain

01 Vector Database – ChromaDB

02 Evaluators – String Evaluators

03 Large Language Model – Ollama Family

04 User Interface – Streamlit



Ollama

DATA SOURCES

PDF'S regarding the:

- 01 Portuguese Penal Code
- 02 Portuguese Processual Penal Code
- 03 Practical Cases and respective Answers
- 04 Notes taken from Portuguese Law Classes



DATA EXTRACTION

EXTRACTION TOOLS

Py2PDF

- Detailed Customization
- Limited features for extracting data
- Only suitable to extract data from images

PDFPlumber

- Handles Complex Layouts
- High Accuracy in text and table data extraction
- Can handle both Scanned and Digital PDF'S

Tesseract

- Requires image pre-processing
- More Suitable to image based PDF's
- Poor performance on PDF's with Complex Layouts

DATA PROCESSING

01 Data Processing of TXT Files

01 Clean the Text

02 Character Normalization

03 Tokenization

04 Indentation



DATABASE

CHROMADB

Storing PDF's and Vector Embeddings

- 01** Stores the PDF's as soon as it initializes
- 02** In each question, the Retriever provides Documents for extra context along with the User Input
- 03** In each response given by the LLM, it stores the Answer in a Vector Embedding



ChromaDB

EXAMPLE

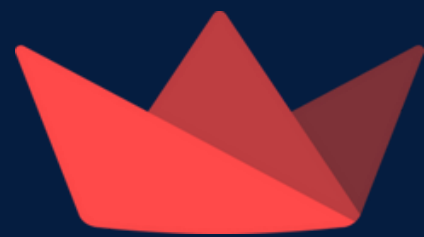
Question:

As finalidades do processo penal podem ser entendidas em que sentido?



Answer:

*As finalidades do processo penal podem ser entendidas em dois sentidos: o **sentido axiológico** e o **sentido sociológico**.*



Streamlit

GRAPHICAL INTERFACE

×



Settings

☒ Reload Embeddings

LLM Model Name

llama2

mistral

llama2

zephyr

Deploy

⋮

 Em que casos se podem aplicar os artigos do Livro 1, Título 4 do Código Penal Português ?

 Com base na pesquisa fornecida, os artigos do Livro 1, Título 4 do Código Penal Português podem ser aplicados nos seguintes casos:

1. Infrações puníveis por lei: A vítima ou seus sucessores têm o direito de apresentar uma queixa privada por crimes puníveis por lei, como assassinato, roubo, danos à propriedade, difamação e outros.

2. Difamação: Indivíduos que foram difamados também podem apresentar uma reclamação privada nos termos do Artigo 149, buscando indenização por danos causados à sua reputação. Portanto, esses artigos podem ser aplicados em casos de assassinato, roubo, danos à propriedade, difamação e outros crimes. Esta é uma questão importante para garantir que as vítimas e as partes afetadas tenham um meio de buscar justiça nos casos em que os promotores públicos podem não priorizá-los ou podem ser incapazes de fazê-lo devido a limitações de recursos ou outros motivos.

Envia Mensagem ao LawTalk

>

BENCHMARKING

Accuracy and Precision Measurements

- 01 Use of an Evaluator from LangChain
- 02 Creation of a Test Dataset with question and expected answer
- 03 Creation of Quantity and Quality Criteria
 - Scores between 1 and 10
 - With each score comes a quality evaluation
 - Quality evaluation based on accuracy and precision



LangChain

RESULT ANALYSIS

Mistral

- Scores between 1 and 10
- Extremely inconsistent with the answers
- Susceptible to hallucinations
- Answer Time : 25 to 35 secs

Llama2

- Scores between 7 and 10
- Very consistent with the answers
- Not susceptible to hallucinations
- Answer Time : 30 to 40 secs

Zephyr

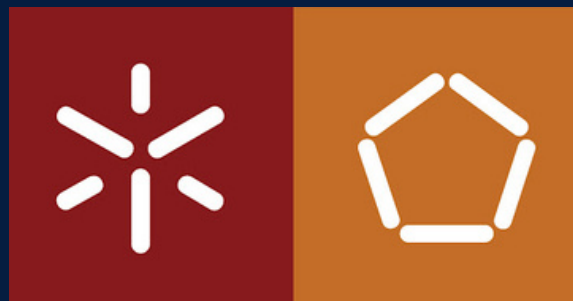
- Scores between 5 and 10
- Mostly consistent with the answers
- Not susceptible to hallucinations
- Answer Time : 35 to 45 secs



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